

# CS 677: Distributed and Operating Systems

## Lab 3: Evaluation Document (Part 4)

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### Requirement:

Run 5 clients on your local machine. Measure the latency seen by each client for different types of requests. Change the probability  $p$  of a follow up trade request from 0 to 80%, with an increment of 20%, and record the result for each  $p$  setting. Also do the same experiments but with caching turned off, estimate how much benefits does caching provide by comparing the results.

### Evaluation Method:

To conduct this experiment, we set the total number of requests to 1000 to obtain accurate measurements. We passed the public IP address of the EC2 instance to the Dockerfile.client and varied the probability as specified. For each probability value, we execute:

```
docker-compose -f 'docker-compose-client.yml' up -d --build 'clientforloadtesting'
```

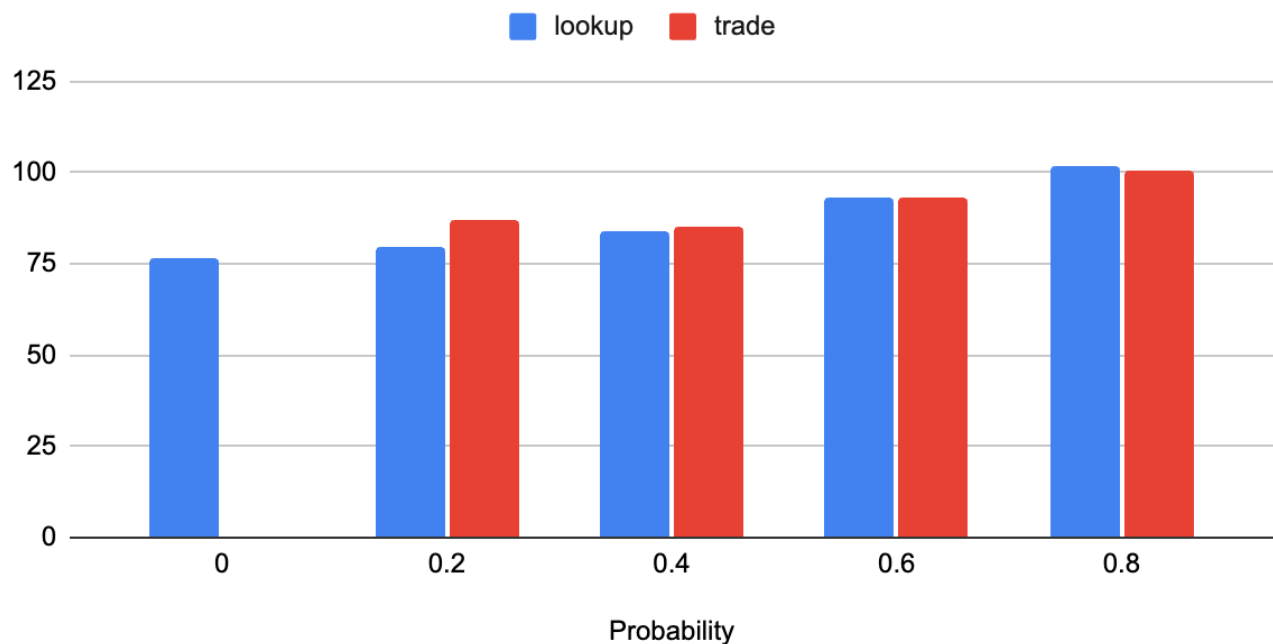
And collect latency measurements from each of the 5 clients, and then computed the average.

The experiment was repeated with caching enabled and disabled to evaluate the performance impact. The following screenshot shows sample measurements for a client when the probability was set to 0.6.

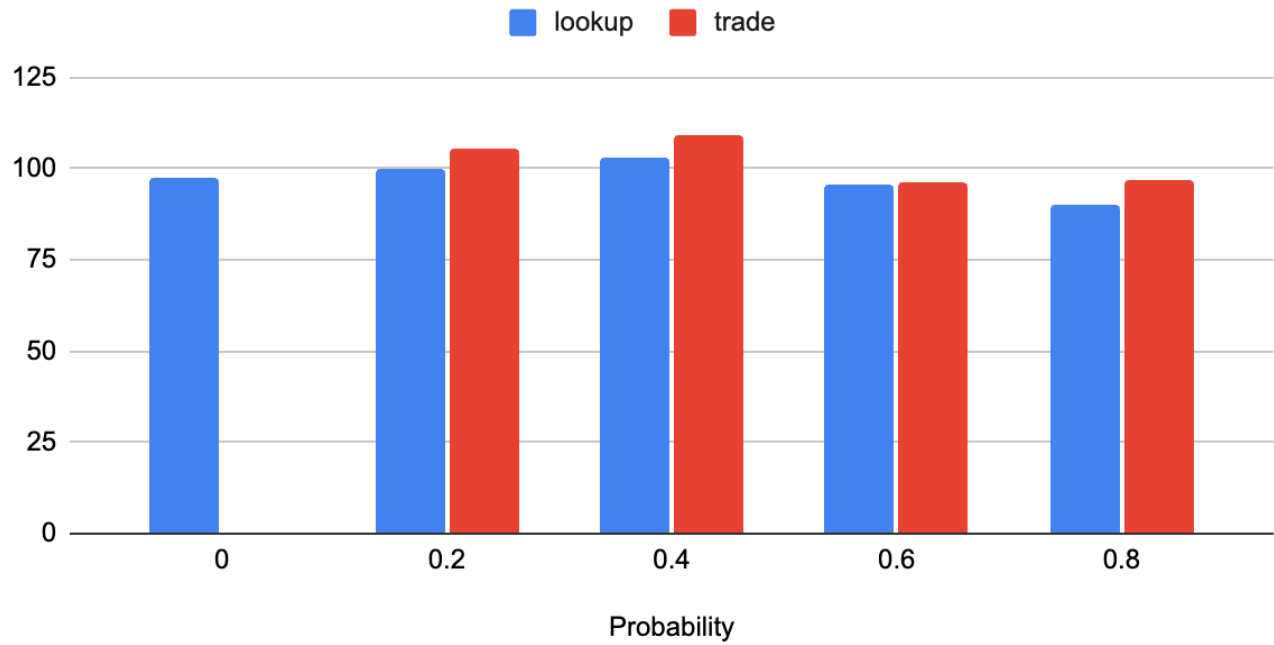
```
2025-05-03 16:34:43 Server response matches with the locally stored information for the transaction number 1535
2025-05-03 16:34:43 Order Lookup:
2025-05-03 16:34:43 /orders/1535
2025-05-03 16:34:43 Server's Response: {"data":{"name":"Google","quantity":1,"type":"buy"}}
2025-05-03 16:34:43 Server response matches with the locally stored information for the transaction number 1535
2025-05-03 16:34:43
2025-05-03 16:34:43 Average Lookup Latency: 57.75 ms (based on 649 lookup requests)
2025-05-03 16:34:43 Average Trade Latency: 72.85 ms (based on 349 trade requests)
```

### Data:

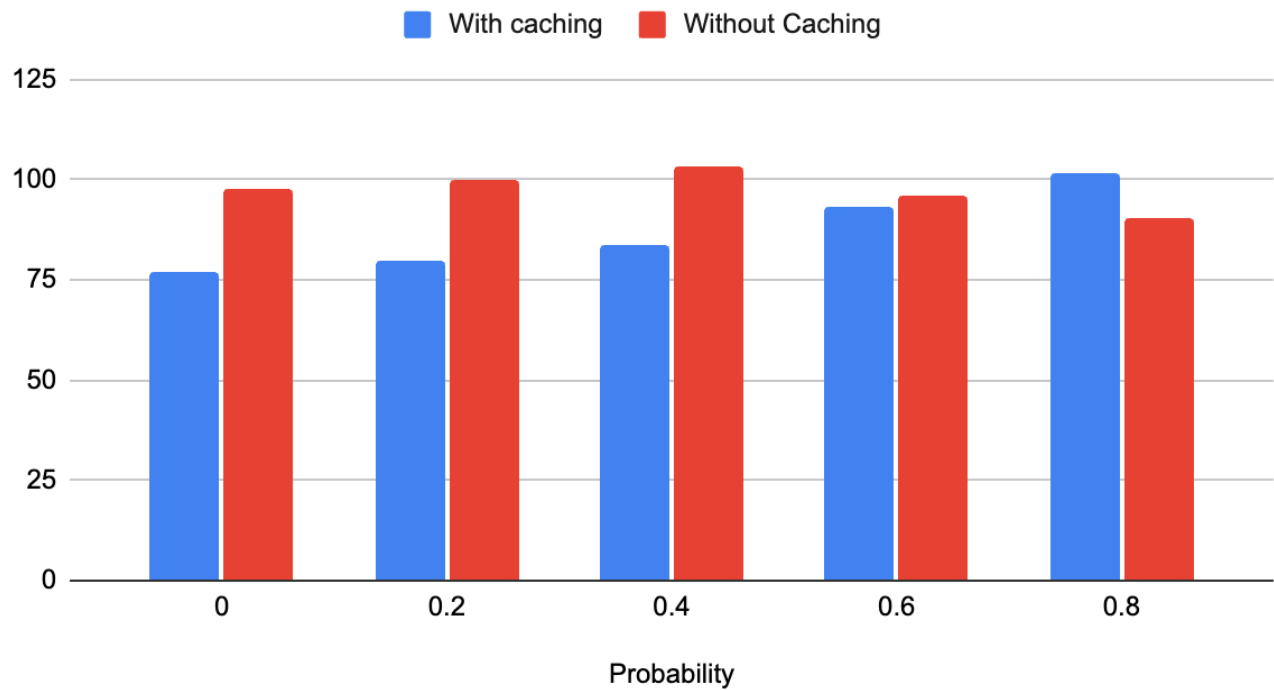
Average latency of lookup and trade requests in milliseconds (when caching is enabled)



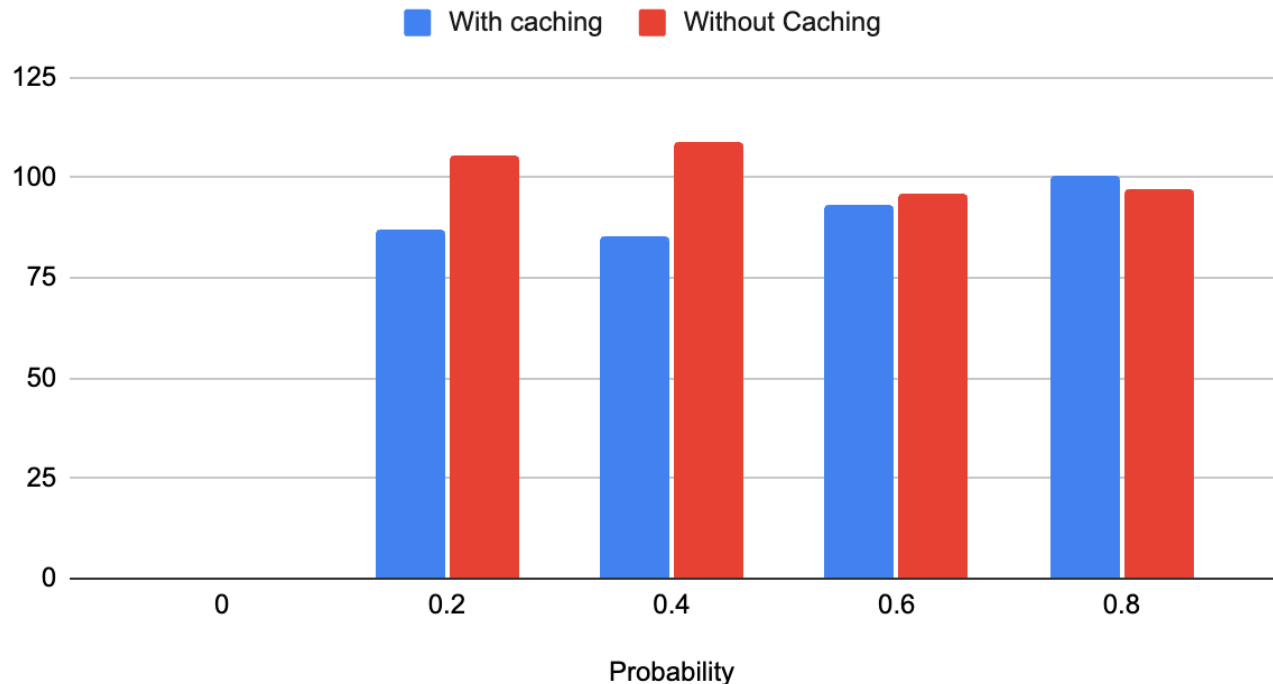
Average latency of lookup and trade requests in milliseconds  
(when caching is disabled)



Average latency of lookup requests (in ms)



## Average latency of trade requests (in ms)



### **Explanation:**

Enabling caching clearly provides a benefit below probability of 0.6 for both request types. At 0.6 and above, the requests without caching are either about equal or slightly lower in latency than requests with caching.

**Below 0.6 Probability:** We get many more lookup requests than orders.

With caching, this means that there are less orders, which means less invalidation locking the cache in the frontend. Additionally, we get more of a benefit from cache utilization with more lookup requests coming in.

In addition to improving the performance of lookup requests, caching also has a positive impact on trade request performance. When the order service receives a trade request, it sends a query to the catalog to verify whether the trade can be fulfilled. However, if another client has acquired a read lock on the catalog around the same time, the service must wait until that lock is released before completing the trade request, which introduces latency.

Without caching, lookup requests do not get the benefit of cache utilization but we also do not have threads waiting for locks at the cache. At low probability, less orders means less cache invalidation and less lock time on the cache but the benefit gained from low lock waiting time does not outweigh the lost benefit of caching.

**Above 0.6 Probability:** We get more order requests, but not more orders than lookups.

With caching, more orders means more cache invalidation. Less lookup requests means lower cache utilization. The combination of these factors means that the cache is not utilized very often and when it is, it has a higher chance of being locked, leading to further delay.

Additionally, order requests cannot be completed until the catalog service can get a lock on the cache to invalidate for a successful trade. Order requests already compete for locks in the catalog, so to also compete for locks in the cache (due to invalidation from successful trades), while there are more trades occurring due to higher probability, leads to higher latency.

### **Cache Replacement and Crash Failure Simulation:**

**Note:** The detailed answers to the following questions are not included in this evaluation document. Instead, they are provided in the output.pdf file.

During office hours, we confirmed that it is acceptable to place the answers to these Part 4 questions either in the evaluation document or in the output file. We chose to include them in output.pdf for clarity and better presentation via images of the runtime behavior and logs.